Training Small LLMs to Solve Systems of Equations by a Specified Method

Verified data, solution format, preference tuning, and synthetic scaling

Hassan Raza Shaikh · Preference Learning in Education

October 2026 (revised)

Changes since the previous draft. (1) All results now carry 95% confidence intervals (cluster bootstrap over the 26 test systems). (2) Two new controls were run: a compute-matched control (1500 iterations) and a distribution-matched synthetic set. (3) Two earlier claims are revised: the worked-vs-terse format effect is statistically established for the general model only; and the earlier explanation of the scaling result ("more iterations") was wrong — longer training does not hurt, and synthetic data did not help even when distribution-matched. (4) An integrity bug in a sympy simplification routine was found and fixed; an audit showed it affected none of the data used in any result.

1. Objective

Build a verified dataset of math problems paired with **multiple solution methods**, and test how to fine-tune a small language model to solve a linear system by a *specified* method (substitution, elimination, Cramer's rule, matrix inverse, Gaussian elimination, graphing). Controlled question: what moves accuracy — the base model's prior, the **format** of the training targets, the training stage (SFT vs. DPO), training length, or added synthetic data?

2. Data and verified pipeline

- **Source:** `CAHLR/OATutor-Content` (13,287 problems). The systems-of-equations slice has 751 problems; **164** state a parseable, uniquely-solvable linear system in text.
- **Questions are real; only solutions are generated.** Every solution is computed with **sympy** and verified by substituting the answer back into the original equations. No language model writes the maths, so no hallucinated arithmetic enters the data.
- **Two target formats:** *terse* (compact steps) and *worked* (every computation shown, e.g. $3(3y) - 4y = 5 \Rightarrow 5y = 5$).
- **Preference pairs (DPO):** verified-correct vs. deterministically corrupted, verified-wrong.
- **Synthetic systems** (training only): an original set, and a *distribution-matched* set whose coefficients and constants are bootstrapped from the real **training** split, matching it on 2/3-variable mix (0.66), small coefficients (0.73 vs 0.71), fractional coefficients (0.06) and fractional solutions (0.38). Both sets: 5,624 examples.
- **Verification:** independent re-check, **0 failures** across all real and synthetic solutions and preference pairs; displayed steps are additionally screened for irrational rendering artifacts (none).

3. Experimental setup

- **Split by system:** 164 → 114 train / 24 valid / **26 test** (126 method-solutions), no leakage; the identical test set is used in every condition.
- **Matched base models (4-bit, MLX, Apple Silicon):** `Qwen2.5-1.5B-Instruct` (general) and `Qwen2.5-Math-1.5B-Instruct` (math). LoRA (8 layers) via `mlx-lm-lora`.
- **Conditions:** SFT at 600 iterations (terse, worked), DPO on top of SFT ($\beta = 0.1$), and at 1500 iterations: worked SFT on the 114 real systems (compute-matched control), plus the same with 5,624 original or distribution-matched synthetic examples added.
- **Metric:** exact answer accuracy, graded by sympy. **95% CIs** from a cluster bootstrap that resamples the 26 test *systems* (items within a system are not independent); contrasts use paired resampling (B = 10,000).

4. Results

Condition	general	math
Base (no training)	0.389 [0.27, 0.50]	0.381 [0.29, 0.46]
Terse SFT (600 it)	0.087 [0.04, 0.15]	0.214 [0.11, 0.32]
Terse SFT + DPO	0.016 [0.00, 0.04]	0.135 [0.08, 0.19]
Worked SFT (600 it)	0.159 [0.09, 0.23]	0.286 [0.16, 0.41]
Worked SFT + DPO	0.000 [0.00, 0.00]	0.111 [0.05, 0.17]
Worked SFT (1500 it), real only	**0.183** [0.09, 0.27]	**0.389** [0.25, 0.52]
+ original synthetic	0.095 [0.04, 0.15]	0.270 [0.17, 0.37]
+ distribution-matched synth.	0.143 [0.08, 0.21]	0.238 [0.15, 0.33]

Table 1: Exact answer accuracy on the 26 held-out real systems, with 95% cluster-bootstrap CIs.

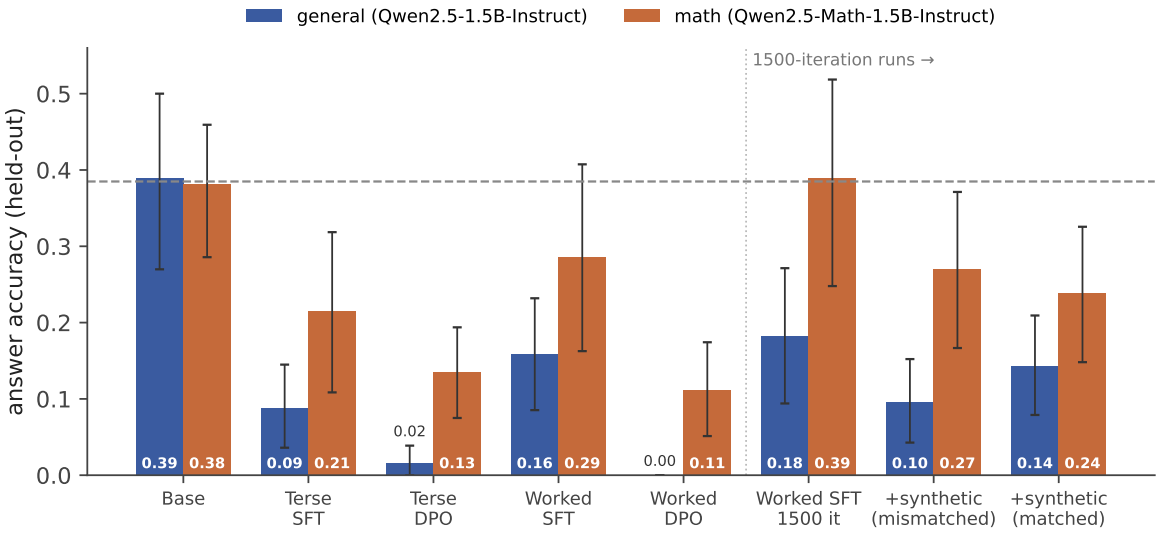


Figure 1: Accuracy by condition with 95% CIs. Runs right of the dotted line use 1500 iterations; the dashed line is the shared base level.

Contrast (difference in accuracy)	general	math
Format: worked − terse (SFT, 600 it)	**+0.071** [+0.02, +0.13]	+0.071 [−0.01, +0.16]
DPO − SFT (terse)	**−0.071** [−0.12, −0.02]	**−0.079** [−0.16, −0.01]
DPO − SFT (worked)	**−0.159** [−0.23, −0.09]	**−0.175** [−0.30, −0.07]
Length: 1500 − 600 it (worked)	+0.024 [−0.04, +0.09]	**+0.103** [+0.03, +0.17]
Add original synthetic (equal compute)	**−0.087** [−0.17, −0.01]	**−0.119** [−0.22, −0.02]
Add matched synthetic (equal compute)	−0.040 [−0.13, +0.04]	**−0.151** [−0.28, −0.03]
Matched − original synthetic	**+0.048** [+0.02, +0.08]	−0.032 [−0.11, +0.04]
Best SFT (1500 it) − base	**−0.206** [−0.31, −0.10]	+0.008 [−0.10, +0.12]

Table 2: Paired contrasts with 95% CIs. Bold: interval excludes zero.

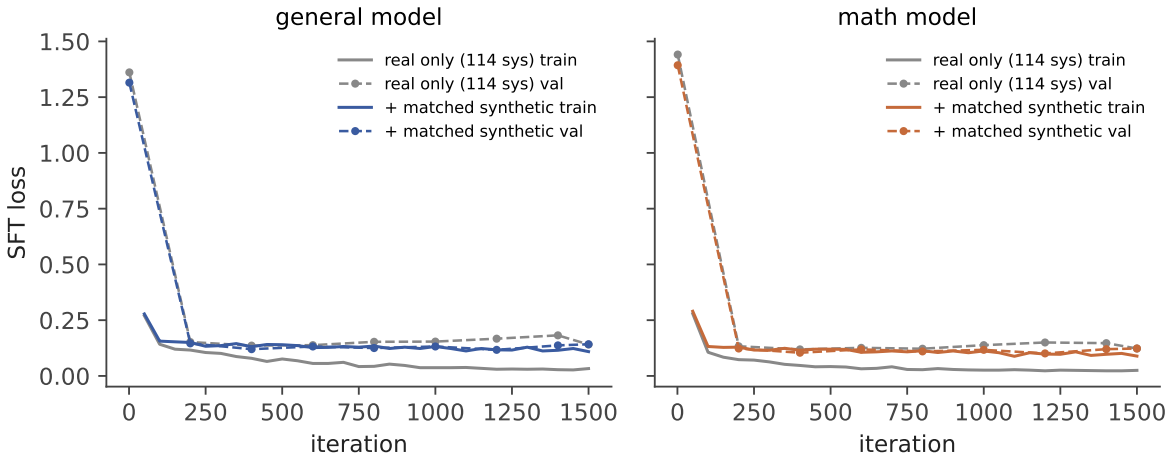


Figure 2: SFT loss at 1500 iterations (train solid, validation dashed; validation = real held-out systems). With real data only, training loss falls to ~0.03 while validation loss drifts up (memorisation). Adding matched synthetic data keeps validation loss flat and lower — yet exact-answer accuracy does not improve (general) or falls (math).

Mechanism of the terse failure (observed in outputs). After terse SFT the model imitates the format perfectly but fabricates the intermediate step: on $2x - 6y = 0,\ 3x - 4y = 5$ it writes $x = 3y$ then invents $15 - 18y = 0$ (should be $5y = 5$). Worked targets show that step.

5. Findings

1. **DPO hurts at this scale — the most robust result.** All four DPO-vs-SFT contrasts are significantly negative (−0.07 to −0.18), even though the preference pairs are perfectly separable by a rule-based judge.
2. **Worked targets beat terse targets** for the general model (+0.071, significant). The math model shows the same point estimate, but its interval includes zero.
3. **Longer training helps the math model** (+0.103, significant) and brings it to parity with its base (0.389 vs 0.381). The general model stays significantly below its base.
4. **Synthetic scaling did not help.** At equal compute, adding the original synthetic data hurt both models. Distribution matching on four summary statistics removed the harm for the

general model (matched vs original +0.048, significant) but not for the math model (matched synthetic still −0.151 vs real-only).

5. **Validation loss is a misleading signal here.** Synthetic data lowered loss on real held-out solutions while exact-answer accuracy stayed flat or fell.

6. Interpretation and next steps

The following are **hypotheses, not yet tested**:

- *Joint structure.* Matching marginal statistics is not enough; curated textbook systems are built so intermediate arithmetic stays clean, while independently sampled coefficients are not. Test: constrain synthetic systems to integer intermediate values.
- *Dilution.* Synthetic examples are 91% of the scaled training set. Test: vary the real:synthetic ratio / upsample real systems.
- *Loss vs. correctness.* Token loss is dominated by template text; arithmetic tokens are few. Test: select checkpoints by sympy accuracy on the validation set rather than by loss.
- *Method-faithfulness* (does the model use the requested method) is logged but currently keyword-based; a real check is the next metric to build.
- *Scale.* All results are for 1.5B/4-bit models; repeat at 7B when a GPU is available.

Limits. 26 test systems give wide intervals (roughly ±0.1); differences smaller than that should not be over-read. Each condition is a single training run (one seed).

Reproducible pipeline: questions verbatim from OATutor; solutions and grading by sympy; every target independently re-verified (0 failures). Code, configs, logs and per-run results:\ `github.com/Hassan-Raza-Shaikh/Preference-Learning-in-Education`